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LiaScript as an AI-Ready Authoring Format

Comparing Skill-Based and Agentic Course Generation

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LiaScript courses are plain text: generative AI can write them, and authors can still read and edit every line. We compare two ways to get there — a lightweight **skill** and a **spec-driven multi-agent workflow.**

1 Why LiaScript fits AI generation

Markdown-near	a Markdown superset, a format LLMs handle well
Locally interpretable	one text file, rendered in the browser, no build step
Versionable	plain text, Git-friendly, unlike LMS packages
Compact	core syntax fits into a context window as a skill

```
# Paint estimate                          ← heading = page

--{{0}}--                                 ← read aloud (text-to-speech)
A wall is 5 m wide and 2.5 m high.

How many liters at 0.2 l/m²?

[[2.5]]                                   ← text quiz, auto-checked

Which coat goes on first?

- [(X)] Primer                            ← single-choice quiz
- [( )] Top coat
```

Try it live in the LiaScript LiveEditor

DiAgnostiK — Lehren und Lernen im Handwerk

Project website uelu-digital.de

Digital unterstützte überbetriebliche Lehrlingsunterweisung im Handwerk

An ESF Plus project of TU Bergakademie Freiberg, Geokompetenzzentrum Freiberg e. V. and Handwerkskammer Dresden.

Inter-company apprenticeship training (ÜLU) in five trades — painters, SHK plant mechanics, joiners, dental technicians, interior decorators — gets accompanying competence assessment, so that training gaps are detected early.

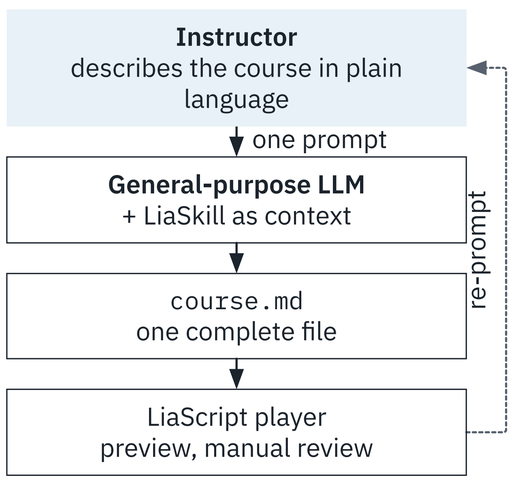
One building block is an **AI-based task generator.** Trainers are subject experts, not markup experts: exactly the gap this work addresses.

2 Two process models: Skill vs. Multi-Agent

Same grammar — who acts, which task, which artifact — at different granularity.

A · Skill-based

One prompt, one file.

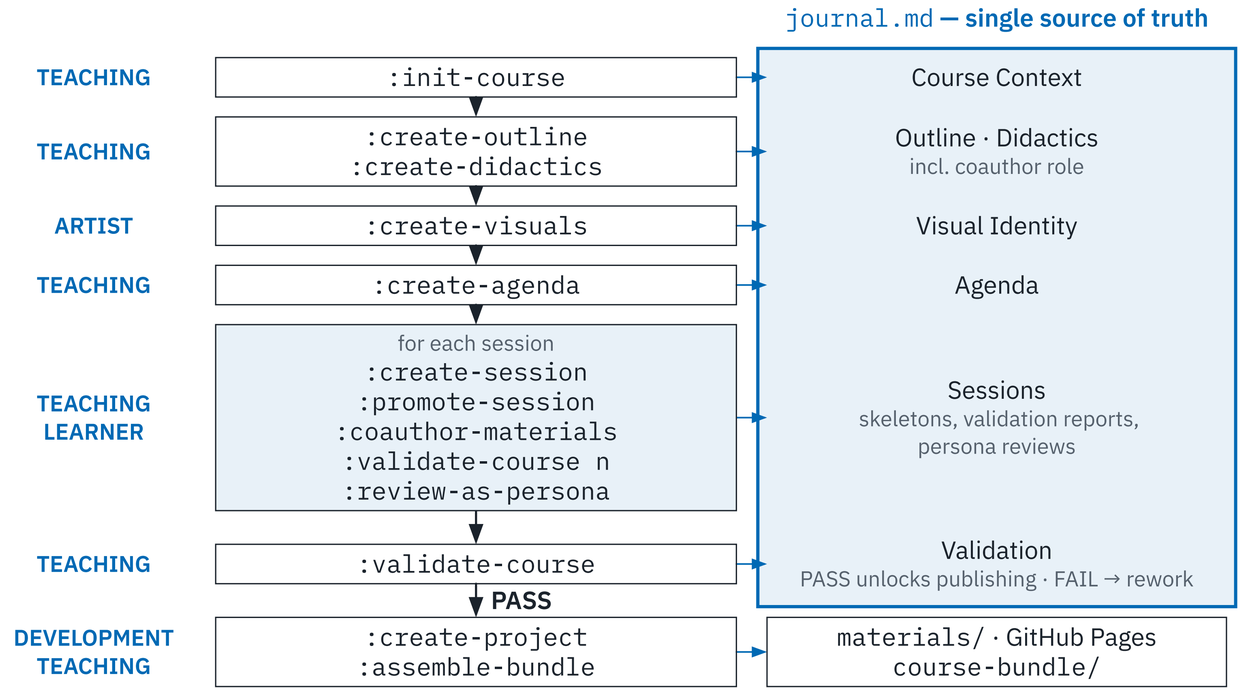


Syntax knowledge: LiaSkill — `SKILL.md` + `references/` (advanced, templates, voices)

LiaSkill github.com/LiaScript/LiaSkill

B · Spec-driven multi-agent

Teaching-Agent, built on the BMAD method: specify first, then generate.



Syntax knowledge: `liascript-cheat-sheet.md` and templates, loaded by the tasks that need them — not the skill from A.

Context is not re-prompted each time — it is defined as an artifact and shared by all agents.

Teaching-Agent github.com/LiaScript/teaching-agent

Both approaches solve the same subproblem — getting LiaScript syntax into the model — at different granularity. **The constant is not the skill, but LiaScript itself.**

3 Comparison

Dimension	A · Skill	B · Multi-Agent
Entry barrier	very low	medium
Artifact scope	single file / learning unit	complete course
Didactic control	implicit in the prompt	explicit (outline, didactics, coauthor role)
Traceability	low	high (`journal.md`, reports, versions)
Quality assurance	manual review	checklists, persona review, publishing gate
Syntax transfer	`SKILL.md` + references	cheat sheet and templates as dependencies
Interaction model	one prompt → file	commands, dialogue, IDE
Suitable for	quick single contents	complex, collaborative courses

4 Limitations

Neither approach removes the need for expert review. **A** has no check across units: factually wrong but syntactically valid content can go undetected. **B**'s PASS certifies structural and didactic consistency, not factual correctness. Both inherit the model's error rate **— not for unattended, high-stakes use.**

5 At the demo

Both prototypes live: one prompt → one file, and the full workflow with dashboard, validation report and persona review.

TU Bergakademie Freiberg · TUBAF Die Ressourcenuniversität. Seit 1765. · GKZ Freiberg · Handwerkskammer Dresden

Video: Teaching-Agent in action

Talk: "Ein Prompt ist kein Team"

Kofinanziert von der Europäischen Union

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BMAD method: github.com/bmad-code-org/BMAD-METHOD · Spec Kit: github.com/github/spec-kit · LiaScript: liascript.github.io

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